

BITS PILANI, DUBAI CAMPUS
DUBAI INTERNATIONAL ACADEMIC CITY, DUBAI

COURSE	: COMPUTER PROGRAMMING (CS F111)
COMPONENT	Practical Week 6: 1D Arrays

Q1 Write a C program to read 5 subjects marks of a student and calculate his/her Total and Percentage using arrays. And print the result as following criteria Calculate percentage and grade according to following:

Percentage $\geq 90\%$: Grade A

Percentage $\geq 80\%$: Grade B

Percentage $\geq 70\%$: Grade C

Percentage $\geq 60\%$: Grade D

Percentage $\geq 40\%$: Grade E

Percentage $< 40\%$: Grade F

Answer:

Code:please scroll down,

```

1  #include <stdio.h>
2
3  int main() {
4      int marks[5], i;
5      int total = 0;
6      float percentage;
7
8      printf("Enter marks of 5 subjects:\n");
9      for(i = 0; i < 5; i++) {
10         scanf("%d", &marks[i]);
11         total += marks[i];
12     }
13
14     percentage = total / 5.0;
15
16     printf("\nTotal = %d\n", total);
17     printf("Percentage = %.2f%%\n", percentage);
18
19     if(percentge >= 90)
20         printf("Grade = A\n");
21     else if(percentge >= 80)
22         printf("Grade = B\n");
23     else if(percentge >= 70)
24         printf("Grade = C\n");
25     else if(percentge >= 60)
26         printf("Grade = D\n");
27     else if(percentge >= 40)
28         printf("Grade = E\n");
29     else
30         printf("Grade = F\n");
31
32     return 0;
33 }

```

Input:

Run

85 90 78 88 92

Output:

Your Output

```

Enter marks of 5 subjects:

Total = 433
Percentage = 86.60%
Grade = B

```

Q2. Write a C program to read two mark lists of equal size (10 size) and check whether they are identical or not.

E.g.

mark list 1 : 22 33 44 55 66

mark list 2 : 22 33 44 55 66

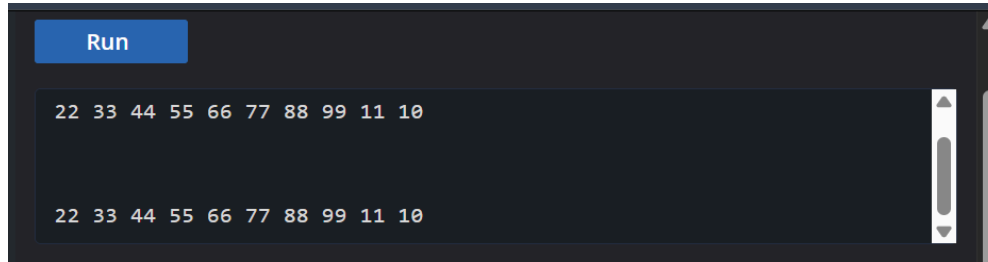
Output : Identical

Answer:

Code:

```
1  #include <stdio.h>
2
3  int main() {
4      int a[10], b[10];
5      int i, flag = 1;
6
7      printf("Enter 10 elements for first array:\n");
8      for(i = 0; i < 10; i++) {
9          scanf("%d", &a[i]);
10     }
11
12     printf("Enter 10 elements for second array:\n");
13     for(i = 0; i < 10; i++) {
14         scanf("%d", &b[i]);
15     }
16
17     for(i = 0; i < 10; i++) {
18         if(a[i] != b[i]) {
19             flag = 0;
20             break;
21         }
22     }
23
24     if(flag == 1)
25         printf("Output: Identical\n");
26     else
27         printf("Output: Not Identical\n");
28
29     return 0;
30 }
```

Input:

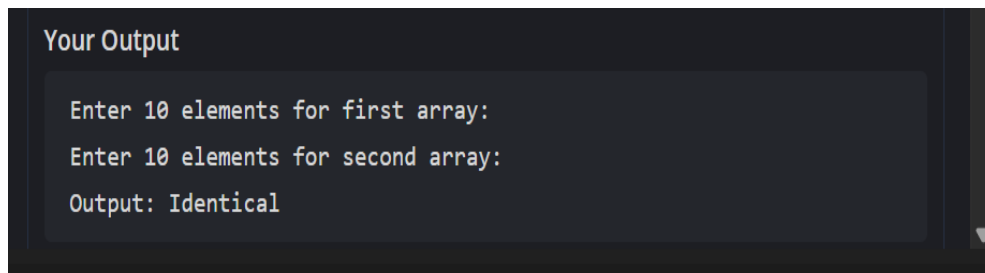
A screenshot of a code editor with a dark background. At the top left, there is a blue button labeled "Run". Below it, there are two lines of text, each containing the sequence of numbers "22 33 44 55 66 77 88 99 11 10". The text is white and appears to be input or output data.

```
Run

22 33 44 55 66 77 88 99 11 10

22 33 44 55 66 77 88 99 11 10
```

Output:

A screenshot of a code editor with a dark background. The text "Your Output" is at the top left. Below it, there are three lines of text: "Enter 10 elements for first array:", "Enter 10 elements for second array:", and "Output: Identical". The text is white and appears to be the output of a program.

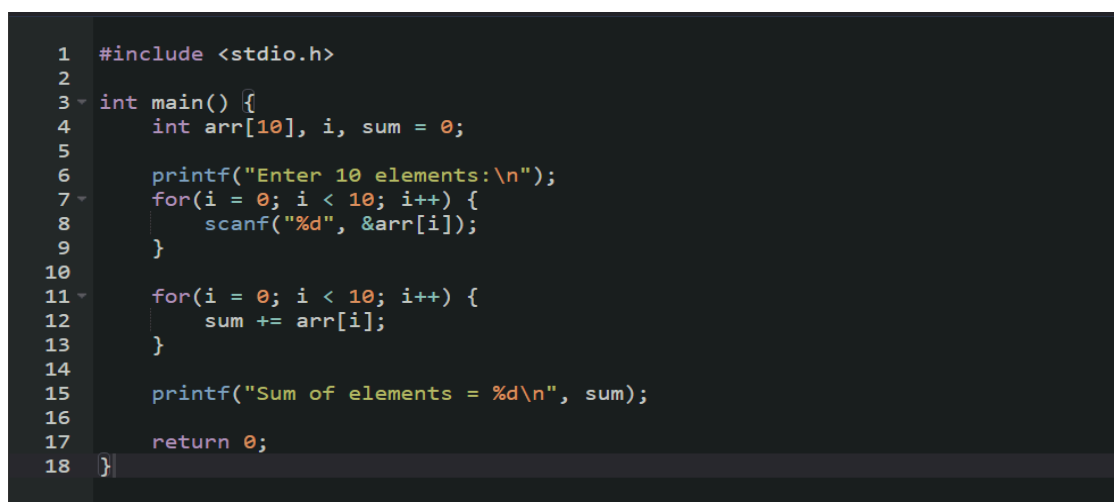
```
Your Output

Enter 10 elements for first array:
Enter 10 elements for second array:
Output: Identical
```

Q3. WAP to find the sum of all elements in an integer array.

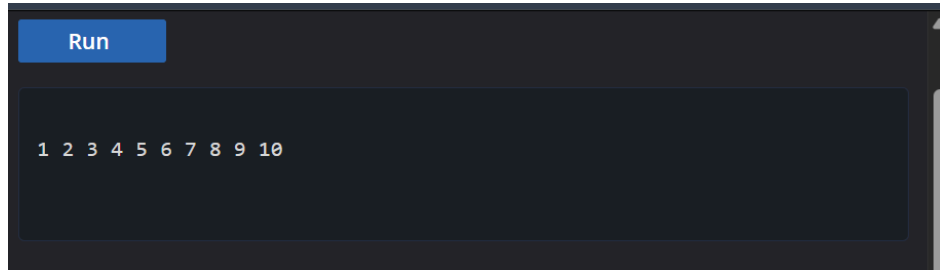
Answer:

Code:

A screenshot of a code editor with a dark background. It contains C code to find the sum of elements in an integer array. The code is numbered from 1 to 18 on the left side. The code includes a header file, declares an array and variables, prompts for input, reads 10 elements, calculates the sum, and prints the result.

```
1  #include <stdio.h>
2
3  int main() {
4      int arr[10], i, sum = 0;
5
6      printf("Enter 10 elements:\n");
7      for(i = 0; i < 10; i++) {
8          scanf("%d", &arr[i]);
9      }
10
11     for(i = 0; i < 10; i++) {
12         sum += arr[i];
13     }
14
15     printf("Sum of elements = %d\n", sum);
16
17     return 0;
18 }
```

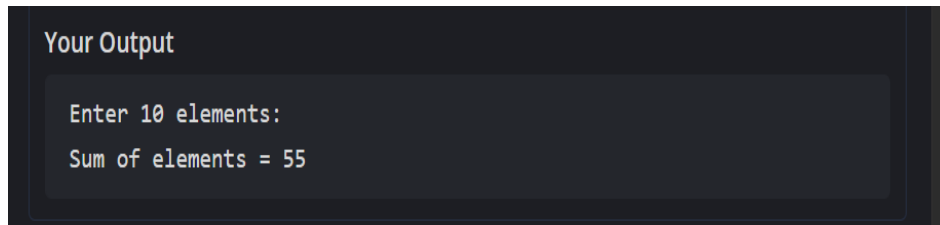
Input:



```
Run

1 2 3 4 5 6 7 8 9 10
```

Output:



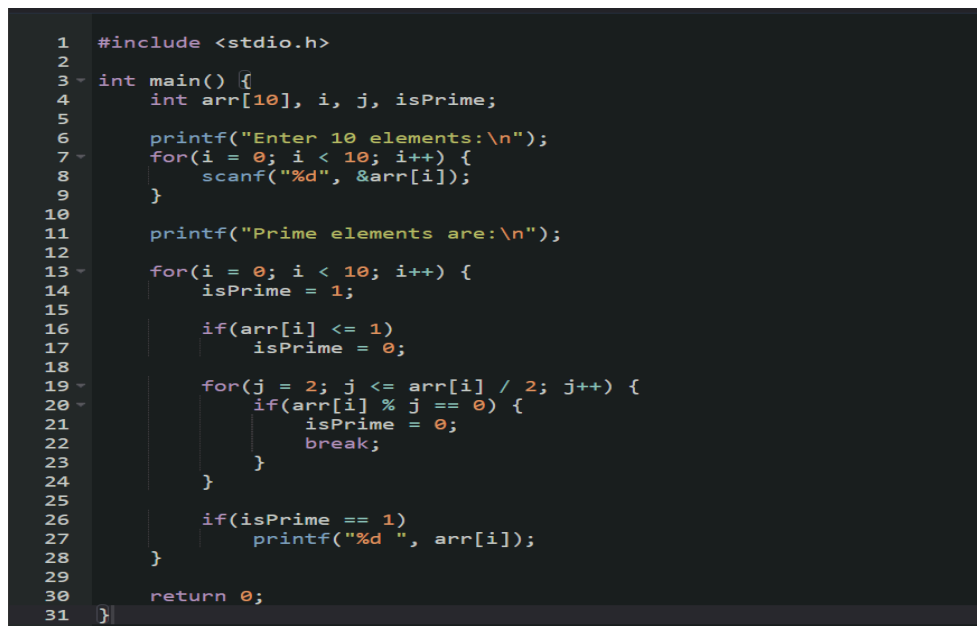
```
Your Output

Enter 10 elements:
Sum of elements = 55
```

Q4. WAP to find the all the prime elements in an array.

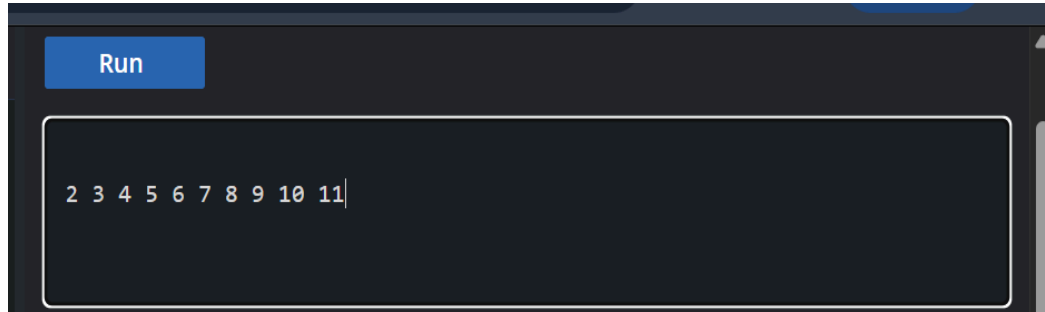
Answer:

Code:



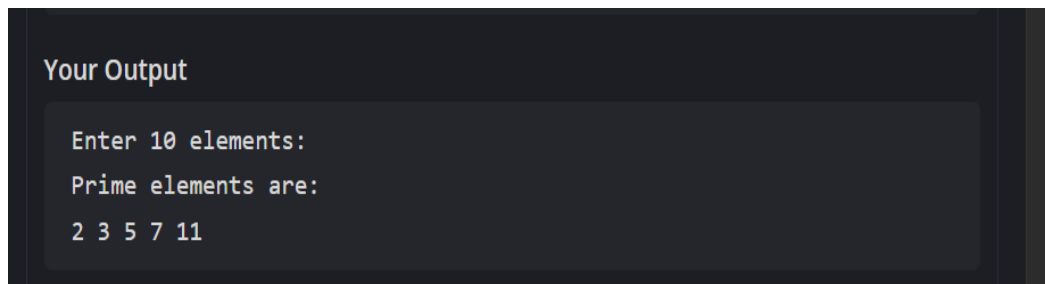
```
1  #include <stdio.h>
2
3  int main() {
4      int arr[10], i, j, isPrime;
5
6      printf("Enter 10 elements:\n");
7      for(i = 0; i < 10; i++) {
8          scanf("%d", &arr[i]);
9      }
10
11     printf("Prime elements are:\n");
12
13     for(i = 0; i < 10; i++) {
14         isPrime = 1;
15
16         if(arr[i] <= 1)
17             isPrime = 0;
18
19         for(j = 2; j <= arr[i] / 2; j++) {
20             if(arr[i] % j == 0) {
21                 isPrime = 0;
22                 break;
23             }
24         }
25
26         if(isPrime == 1)
27             printf("%d ", arr[i]);
28     }
29
30     return 0;
31 }
```

Input:



A screenshot of a code editor interface. At the top left, there is a blue button labeled "Run". Below it is a large text input field containing the sequence of numbers "2 3 4 5 6 7 8 9 10 11" followed by a cursor. The editor has a dark theme and a scrollbar on the right side.

Output:



A screenshot of a code editor interface showing the output of a program. The text "Your Output" is at the top. Below it, the output is displayed in a light gray box: "Enter 10 elements:", "Prime elements are:", and "2 3 5 7 11". The editor has a dark theme and a scrollbar on the right side.
